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Object Detection for Contraband Identification using Deep Learning Pipelines

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This concept paper is under review by the supervisors.

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Abstract— Security screening in correctional facilities is crucial for preventing the smuggling of prohibited items, which can pose significant risks to safety and order. This research addresses the limitations of manual X-ray inspection by developing automated object detection pipelines. Faster R-CNN was adapted for grayscale X-ray images, incorporating preprocessing techniques such as log2 filtering and augmentations (NoAugment, CLAHE, Gamma, Emboss, GaussianBlur). Experiments showed NoAugment without log2 filtering achieved most of the highest F1-scores (e.g., surgical_implant 0.812, zipper_buttons 0.449), while aggressive augmentations introduced artifacts or detail loss, reducing performance on small objects (many near-zero F1). These findings suggest baseline approaches outperform contrast-heavy augmentations for X-ray anomaly detection, with implications for faster, more reliable security screening in prisons.

Index Terms— Object detection, Faster R-CNN, X-ray imaging, anomaly detection, data augmentation, correctional security

1 Introduction

Security screening plays a very important role in correctional facilities, especially because inmates may attempt to hide or carry prohibited objects on their bodies. These objects can be dangerous, illegal, or simply not allowed inside the facility. Examples include drugs, cellphones, sharp tools, and many other everyday items that become security risks in this environment. To support officers with this task, full-body X-ray scanning systems allow the human body to be inspected through radiographic images.

Although these scanners provide detailed images, manual inspection is still difficult. Objects often overlap with bones, soft tissue, and clothing. Some materials appear very faint in X-ray images, while others have very similar shapes. Small objects such as piercings, screws, or drugs are especially hard to detect. Because of this, automated object detection using artificial intelligence has become increasingly important for security applications [1, 2].

In recent years, deep learning has achieved strong performance in object detection tasks. Models such as Faster R-CNN [1], YOLO [3], and DETR [4] have shown that computers can learn to locate and classify objects directly from images. These techniques have also been applied to X-ray security imaging [2]. However, full-body X-ray images remain very challenging due to overlapping objects, grayscale appearance, different materials, and a large number of object classes.

This technical paper focuses on building and testing an object detection pipeline for a dataset of full-body X-ray images with 33 different object classes. These classes include not only metallic objects, but also non-metallic items such as clothing, personal belongings, tools, and different forms of contraband.

The goal of this research is to evaluate how well modern deep-learning-based object detection models perform on full-body X-ray images.

The following research questions guide this technical paper:

1. How can Faster R-CNN be adapted for grayscale X-ray data?
2. How do different augmentations and Filter (e.g., CLAHE, Gamma, NoAugment) affect detection performance?
3. How well does the model detect small or low-contrast objects such as jewelry, piercings, drugs, and zipper buttons?
4. What are the main strengths and weaknesses of the detection approach for full-body X-ray images?

These questions are answered through experiments, evaluation metrics, and analysis.

2 State of the Art

Object detection in X-ray security imaging has evolved from hand-crafted features to deep learning-based methods. Early approaches relied on techniques such as HOG and SIFT, which struggled with low contrast, overlapping objects, and grayscale variations typical in X-ray scans.

Modern deep learning models are divided into two main categories: two-stage and one-stage detectors.

2.1 Two-Stage Detectors

Faster R-CNN [1] is a widely used two-stage detector. It combines a Region Proposal Network (RPN) with Fast R-CNN for accurate bounding box detection and classification. The RPN generates candidate regions, while the second stage classifies and refines them. This architecture

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excels in cluttered scenes and small-object detection, making it suitable for contraband identification in X-ray images [2]. Strengths include high accuracy on overlapping objects. Limitations include slower inference and dependence on anchor box design.

2.2 One-Stage Detectors

One-stage detectors such as YOLO [3] perform detection in a single forward pass. These models offer fast inference, suitable for real-time applications. However, they sometimes struggle with very small objects and heavily overlapping items, which are common in full-body X-ray images.

2.3 Deep Learning for X-ray Security Imaging

Deep learning has been applied to X-ray security systems, particularly for baggage and airport screening [2, 5]. Convolutional neural networks successfully detect prohibited objects in complex scans. However, most previous work focuses on baggage rather than full-body inmate scanning, which involves unique challenges like body overlap, grayscale noise, and a large number of classes.

2.4 Gap and Contribution

Existing research often uses synthetic or colored images, while real X-ray challenges—low contrast, heavy occlusion, and class imbalance—remain underexplored. This work addresses that gap by evaluating Faster R-CNN adaptations on a real full-body X-ray dataset, analyzing the impact of contrast enhancement and augmentations on small-object detection. The results provide practical insights for improving security in correctional environments.

3 Materials and Methods

This section describes the dataset, annotation workflow, preprocessing pipeline, data augmentation strategies, and the implementation of the Faster R-CNN model used in this project.

3.1 Dataset

A proprietary dataset of 451 full-body X-ray scans (1216x1500 px, grayscale) was used, sourced from OD Security's THEIA Dataset. Annotations were created in CVAT [6] in Pascal VOC XML format, covering 33 anomaly classes. The dataset was split into 50% training, 30% validation, and 20% testing subsets. Table 1 provides an overview of selected classes and their distribution.

All images are originally grayscale and were converted to three channels to fit the requirements of the backbone network. Each image contains one or more objects belonging to 33 classes. The classes include surgical implant, shoe, dental, zipper, zipper buttons, bag, lighter, cast, glasses, bra, charger, watch, drugs, ankle monitor, padlock, paper, key, cup, piercing, belt, jewelry, cellphone, cuffs, unknown anomaly, prosthetic, screw pocket, collar, buckle, crackpipe, motion artifact, radiopaque, abnormal pose, and unknown definedbyoperator.

For training and evaluation, the dataset was exported from CVAT in Pascal VOC format and organised into a standard directory structure with separate folders for training, validation, and testing.

Table 1. Class distribution and intra-class area variability for the annotated anomaly classes.

Class	Instances	Mean Area	Intra-Class Dist.
zipper_buttons	923	247	189
surgical_implant	662	10488	12181
dental	525	328	407
unknown_anomaly	455	1391	2300
zipper	146	4442	2822
glasses	37	10455	9179
bra	37	13419	5286
buckle	17	1127	1123

3.2 Annotation Workflow

All bounding boxes and labels were created manually using CVAT [6]. Bounding boxes were drawn around every visible object of interest, and each was assigned one of the 33 class labels. In cases where class assignment was ambiguous (e.g., distinguishing between different metal parts or unknown shapes), discussions were held with supervisors or the label "unknown anomaly" or "unknown definedbyoperator" was used. Weekly meetings with OD Security and supervisors provided feedback on annotation consistency and data quality.

3.3 Preprocessing

Preprocessing was applied consistently across all experiments to ensure fair comparison. Images were converted from grayscale to three channels for compatibility with the backbone network. Logarithmic correction $\log_2(1+I)$ was used to compress high-intensity regions and enhance low-intensity anomalies (Figure 1).

Fig. 1. Raw X-ray scan (left) vs. log2-corrected (right).

3.4 Data Augmentation

Due to the limited dataset size, on-the-fly augmentations were applied using Albumentations to improve generalization. Five augmentation strategies were evaluated . Data augmentation strategies evaluated in the experiments (NONE[Baseline], Gaussian Blur, Emboss, Random Gamma, CLAHE).

Stronger augmentations (horizontal/vertical shifts, scaling up to 15%, random brightness/contrast, Gaussian blur, optional CLAHE) were used to increase the variability of the

data set. The baseline model did not use augmentations for controlled comparison.

3.5 Faster R-CNN Implementation

Faster R-CNN was implemented using PyTorch and the torchvision library. The model uses the fasterrcnn_resnet50_fpn_v2 architecture with a ResNet-50 + Feature Pyramid Network (FPN).It made use of ImageNet-pre-trained weights and all backbone layers were trainable.

Training was performed on a GPU with batch size 2, Adam optimizer (learning rate 0.001, weight decay 1e-4), and ReduceLROnPlateau scheduler. Up to 100 epochs were used, with early stopping after 15 epochs without validation improvement. During training, loss was logged to TensorBoard.

Validation was performed every epoch. The model was set to evaluation mode, and validation loss was computed. For the first batch, predictions were visualized with ground truth in TensorBoard. The best model (lowest validation loss) was saved.

Testing used the best checkpoint. Inference was run on the test set, and outputs were decoded into bounding boxes, labels, and confidence scores. Metrics were computed as described in Section 4.

4 Experiments

Two main experiments were conducted to evaluate the impact of the log2 contrast enhancement filter and various data augmentations on Faster R-CNN performance for contraband detection in grayscale X-ray images. The focus was on four key classes: surgical_implant, dental, zipper, and zipper_buttons. These classes are challenging due to small size, low contrast, repetitive textures, and overlap with body structures.

Augmentations (CLAHE, Gamma, Emboss, GaussianBlur) were applied only during training to introduce variability, while validation and testing used clean preprocessing. Early stopping (patience=5 after epoch 10) prevented overfitting by halting training when validation loss plateaued.

Performance was assessed using F1-score, Recall@0.5, and Precision@0.5 at an Intersection over Union (IoU) threshold of 0.5. IoU is defined as:

$$IoU = \frac{\text{Area of Intersection}}{\text{Area of Union}} \tag{1}$$

F1-score balances precision and recall:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

where Precision = TP / (TP + FP) and Recall = TP / (TP + FN). Recall@0.5 and Precision@0.5 were computed at a confidence threshold of 0.5.

The test set contained 846 ground truth objects, with predictions ranging from 1471 to 3874 — higher counts often indicated more false positives from augmentation-induced noise.

4.1 Experiment 1: With Log2 Filter in Preprocessing

The log2 filter enhanced subtle density differences but risked amplifying noise and creating artificial edges when combined with augmentations.

NoAugment (baseline) achieved the highest balanced performance (surgical_implant F1=0.769, dental F1=0.408, zipper F1=0.698, zipper_buttons F1=0.420). Log2 improved edges without distortion.

CLAHE improved recall on larger objects but reduced performance on small/fine items (surgical_implant F1=0.419, dental F1=0.170, zipper F1=0.000, zipper_buttons F1=0.205).

Gamma yielded moderate results (surgical_implant F1=0.289, dental F1=0.254). Exposure adjustments (limit 80-120) were too subtle against log2's aggressive enhancement.

Emboss performed similarly but with higher false positives (surgical_implant F1=0.548, dental F1=0.337). Edge emphasis amplified log2 artifacts, mislabeling zipper buttons.

GaussianBlur performed the worst (surgical_implant F1=0.027, dental F1=0.078, zipper/zipper_buttons near 0). Blurring (limit 7-15) erased critical fine edges in X-rays and causing high false positives and near-zero recall on focused classes.

4.2 Experiment 2: Without Log2 Filter

The log2 filter was removed to test if it contributed to artifacts. Raw X-ray contrast proved more robust.

NoAugment achieved the highest F1-scores (surgical_implant F1=0.812, dental F1=0.437, zipper F1=0.595, zipper_buttons F1=0.449). Without log2, natural densities were preserved, reducing noise for implants and zippers.

CLAHE showed slight gains on surgical_implant (F1=0.600) but remained poor on dental/zippers (many 0s).

Gamma improved compared to Experiment 1 (surgical_implant F1=0.619). Mild adjustments performed better without log2's high contrast.

Emboss was decent (surgical_implant F1=0.641). Edge emphasis helped implants but increased false positives on zippers.

GaussianBlur remained poor (surgical_implant F1=0.483). Blur eliminated critical edges, causing classifier failure on dental/zippers.

Class	F1 (With Log2)	F1 (Without Log2)
surgical_implant	0.769	0.812
dental	0.408	0.437
zipper	0.698	0.595
zipper_buttons	0.420	0.449

Table 2. F1-score comparison for NoAugment across both experiments on focused classes.

All results are stored in the results.pdf for reproducibility.

5 Discussion

The results demonstrate that preprocessing and augmentation choices significantly influence Faster R-CNN performance on

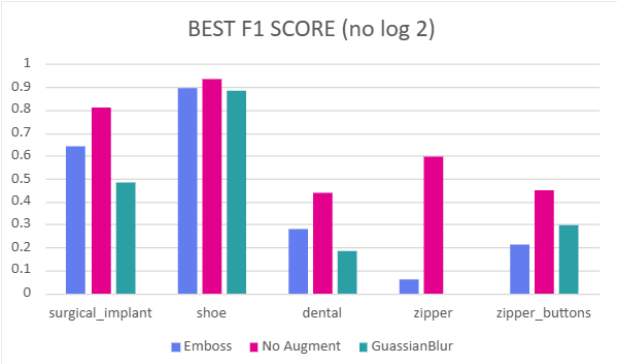

Fig. 2. Best F1-score comparison (Without Log2 Filter) on focused classes. NoAugment outperforms aggressive augmentations, as GuassianBlur creates high-contrast artifacts reducing recall on small objects, While EMBOSS being the best performing Augment but with a lot of False Positives

full-body X-ray images. NoAugment without log2 filtering achieved the highest F1-scores on the focused classes , outperforming all other configurations. This suggests that raw X-ray contrast provides sufficient discriminative features for the ResNet50-FPN backbone, while log2 filtering introduces noise amplification that the Region Proposal Network (RPN) misinterprets as object proposals, increasing false positives and reducing recall on small objects.

Aggressive augmentations such as CLAHE showed mixed outcomes. When combined with log2, CLAHE's tile-based contrast enhancement created artifacts (uneven patches and over-amplified edges), leading to false negatives on small/fine classes (e.g., zipper F1=0.000). Without log2, CLAHE improved slightly on surgical_implant (F1=0.600) but still failed on dental and zippers due to residual edge artifacts. This indicates that high-contrast amplification is detrimental when the baseline intensity range is already limited.

GaussianBlur consistently performed the worst across both experiments (surgical_implant F1 near 0.027–0.483, dental near 0.078–0.175, zipper/zipper_buttons near 0). Blurring smooths out critical fine edges in X-ray images, reducing the backbone's ability to extract discriminative features for small objects. This is particularly harmful for contraband detection, where subtle boundaries (e.g., zippers, implant outlines) are essential for accurate proposal generation.

Gamma and Emboss yielded moderate results, with slight improvements without log2. Gamma's exposure adjustments (limit 80-120) were too subtle to address inherent low-visibility issues, while Emboss's edge emphasis (alpha 0.2-0.5) amplified noise in some cases, increasing false positives on zipper buttons.

The comparison between experiments highlights a clear trend: removing log2 filtering improves precision and recall for NoAugment, as it preserves natural density variations without introducing artificial noise. Aggressive augmentations (CLAHE, GaussianBlur) underperform due to their tendency to distort X-ray-specific features, while the baseline approach (NoAugment without log2) proves more robust for this domain.

Overall, the comparison between experiments shows that preserving natural X-ray intensity distributions is more effective than applying contrast-heavy preprocessing. Simpler preprocessing with minimal augmentation proved more robust for this domain.

Limitations include class imbalance and the relatively small dataset size, which may affect generalization. Nevertheless, the findings provide practical guidance for designing reliable object detection pipelines for correctional X-ray security systems.

6 CONCLUSION

This research evaluated Faster R-CNN adaptations for contraband detection in full-body X-ray images. Experiments demonstrated that NoAugment without log2 filtering achieved the highest performance on key classes (surgical_implant F1=0.812, zipper F1=0.595, zipper_buttons F1=0.449), while aggressive augmentations introduced artifacts or detail loss, reducing detection of small objects.

The results highlight the importance of preserving natural contrast in X-ray data and avoiding over-enhancement. Faster R-CNN's two-stage architecture benefits from clean features, making baseline preprocessing more effective than complex augmentations for this domain.

This work contributes to automated security screening in correctional facilities by providing evidence-based insights into preprocessing and augmentation choices. The findings support the development of more reliable, faster, and scalable anomaly detection pipelines.

7 FUTURE WORK

Future research could explore additional preprocessing techniques, such as adaptive histogram equalization variants or noise reduction filters tailored to X-ray noise characteristics. Expanding the dataset with more diverse scans and rare contraband classes would improve generalization. Hybrid models combining Faster R-CNN with transformer-based elements (e.g., DETR or DEIM) may offer benefits in handling small objects and occlusion. Real-time deployment on edge devices and integration with human-in-the-loop systems are also promising directions.

8 ETHICAL IMPACT STATEMENT

This research adheres to GDPR for data anonymity and uses a proprietary dataset with no personal identifiers. The primary societal benefit is enhanced security screening in correctional facilities, potentially reducing contraband-related risks without introducing bias. Risks include false positives leading to unnecessary inspections, which are mitigated by retaining human oversight. No automated decisions are proposed without human review.

REFERENCES

[1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.

[2] S. Akcay, M. E. Kundegorski, C. G. Willcocks, and T. P. Breckon, "Towards automatic threat detection: A survey of advances of deep learning within x-ray security imaging," *arXiv preprint arXiv:2001.01293*, 2020.

[3] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," *arXiv preprint arXiv:1506.02640*, 2016.

[4] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," *European Conference on Computer Vision (ECCV)*, pp. 213–229, 2020.
[5] Y. Zhao *et al.*, "Object detection in x-ray security imaging: A survey," *Journal of Imaging*, vol. 9, no. 5, p. 98, 2023.
[6] B. Sekachev, N. Zhiltsov, A. Solovyev, A. Trusov, and V. Iglovikov, "Computer vision annotation tool (cvat)," https://github.com/opencv/cvat, 2019.